

Entrepreneurship Bootcamp — Master Cheatsheet

Al-Hussein Technical University (HTU) · Business Track (9 lectures) + Design Thinking Track (7 lectures)

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BUSINESS TRACK

L1 · Innovation & Entrepreneurship

Innovation = creating + implementing new ideas that generate value (economic/social/tech). Not just invention — must create value.

Entrepreneurship = spotting opportunities, mobilizing resources, creating value via new ventures. The 4th factor of production (land, labor, capital, + entrepreneurship).

Innovation Sweet Spot = overlap of 3 circles:

- **Desirability** — people/users want it
- **Feasibility** — technology can build it
- **Viability** — business can sustain it

⚡ All 3 circles must overlap for good innovation.

L1 · Design Thinking Process

Stanford d.school 5 stages:

Empathize → Define → Ideate → Prototype → Test

Starts with **people & needs**, not tech or business models.

Airbnb: found bad host photos → sent photographers → solved sign-up problem.

DT vs Entrepreneurship: DT = **what** to build; Entrepreneurship = **how** to deliver & scale.

L1 · Mindset, Teams & Agility

3Ms: **Mind** (mindset/direction), **Market** (the map of value), **Money** (the fuel — manage wisely).

Growth Mindset: abilities grow with effort; failure = learning. Traits: Curiosity, Vision, Risk Tolerance, Resilience, Adaptability, Persistence.

Business Agility: adapt fast — rapid experimentation, continuous learning, iterative dev, flexible decisions.

High-Performance Teams: collaborate, not just talented.

CATme = peer-feedback system → accountability + spots contribution gaps.

L2 · PEST Analysis

External **macro-environment** scan. Used when entering a new market.

Political	Policies, regulation, stability, licenses, public funding
Economic	Inflation, interest rates, employment, purchasing power, capital access
Social	Culture, demographics, education, lifestyle trends
Technological	Internet/mobile, infrastructure, emerging tech, digital tools

⚡ PEST ≠ PESTEL. PESTEL adds **Environmental** + **Legal** (Design track).

L2 · SWOT Analysis

Strengths	Internal + (what you do well)
Weaknesses	Internal – (what you lack)
Opportunities	External + (trends, gaps)
Threats	External – (competition, regulation)

S+W = **Internal**; O+T = **External**.

⚡ PEST informs the O and T of SWOT.

L3 · Entrepreneurship Approaches

Lean Startup: Build-Measure-Learn loop. **MVP** = smallest build to test a hypothesis. Pivot or persevere on validated learning. *Dropbox*: demo video before coding.

Agile: iterative sprints, continuous delivery + feedback, flexible direction.

Tech-led vs Design-led: "built it because we could" vs "because they needed it." Best = both, but start with human need.

Social Entrepreneurship: social value via business methods. *Grameen Bank microfinance*.

⚡ Lean = eliminate waste/validate fast; Agile = adapt in cycles. They complement, not compete.

L4 · Business Model Canvas (9 blocks)

Right = market/customer · Center = value · Left = operations · Bottom = money. Asks: Desirable? Feasible? Viable?

Customer Segments	Who you serve (demo/geo/behavior/psycho). No segment = no customer.
Value Proposition	Unique value; ties to Jobs/Pains/Gains
Channels	Sales / communication / distribution
Customer Rel.	Personal / Self-service / Automated
Revenue Streams	Sales, fees, subscription, licensing, ads
Key Resources	Physical, intellectual, human, financial
Key Activities	Production, marketing/sales, distribution, service
Key Partners	Suppliers, distributors, alliances, investors
Cost Structure	Fixed + variable; economies of scale/scope

⚡ Mistakes: too many segments (pick a beachhead!), vague VP, unrealistic revenue, ignoring costs.

L5 · Value Proposition Canvas

Customer Profile (right): Jobs (functional/emotional/social), Pains (frustrations/risks), Gains (desired outcomes).

Value Map (left): Products & Services, Pain Relievers, Gain Creators.

Fit = value map matches customer profile.

VP statement: "[Product] helps [customer] achieve [benefit] by [solving pain]." *Uber: fast, reliable, affordable ride at a tap — no waiting/negotiation/cash.*

L5 · Markets & B2B vs B2C

Market = buyers + sellers exchange. Types: consumer, financial, business, online.

B2B	Rational, long cycles, many deciders, ROI-driven
B2C	Emotional, short cycles, personal benefit

⚡ If the user isn't the payer → system game (B2B dynamics).

L5 · Diffusion of Innovation

Innovators 2.5%	Risk-takers, try first
Early Adopters 13.5%	Visionaries, spot benefit early
Early Majority 34%	Pragmatists, risk-averse
Late Majority 34%	Skeptics, price-sensitive
Laggards 16%	Traditionalists, last

⚡ SOM targets Innovators + Early Adopters.

L5 · TAM / SAM / SOM ★

TAM = total demand if 100% adopt (whole market).

SAM = portion you can serve (geo/constraints).

$SAM = TAM \times \text{target segment \%}$

SOM = portion captured short-term (1–3 yrs).

$SOM = SAM \times \text{market share \%}$

Uber Jordan: TAM ~2.4M, SAM ~920K (Amman+Irbid), SOM ~110–120K (12%).

⚡ SOM = your "entry wedge." SAM covers Innovators→Early Majority; SOM = Innovators+Early Adopters.

L6 · Competitor Analysis

Competitor types: Direct (same product+audience), Indirect (diff product, same need), Emerging (disruptors).

Benchmarking → compare vs rivals. **Moon Chart** = radar/spider chart for multi-attribute comparison.

Brand Positioning / Perceptual Map = plot brands on 2 attributes (e.g. Price vs Quality) → reveals market gaps.

USP = Unique + Valuable + Clear. = "what customers want" ∩ "what rivals do poorly."

⚡ Benchmarking shows where you outperform → that becomes your USP.

L6 · Pricing Strategies ★

Cost-Plus	Price = cost + markup %. Ignores willingness to pay.
Value-Based	Price on perceived value; needs differentiation
Competitive	Match rivals (low differentiation)
Penetration	Low entry price → gain share fast → raise later
Skimming	High price first (early adopters) → lower over time. <i>iPhone launch.</i>
Psychological	\$9.99, charm/prestige, bundling, anchoring, scarcity

L7 · 4Ps Marketing Mix

Product	Features, quality, design, branding. Core vs augmented. Lifecycle: Intro→Growth→Maturity→Decline
Price	Signals value, positions brand (see L6)
Place	Distribution: direct / indirect / digital. How + WHERE + WHEN accessed.
Promotion	Advertising, PR, social, content, sales promo. IMC = consistent message.

7Ps (services) adds: **People, Process, Physical Evidence.**

⚡ Each P must align with the others and the target customer.

L8 · Cost, Revenue & Finance

Finance helps: survive, plan, grow, reduce uncertainty. Most startups fail from **financial mismanagement**, not bad ideas.

Mistakes: underestimate costs, overestimate revenue, spend before validating, poor cash flow.

Startup Costs	One-time to launch (equipment, legal, site, licenses)
Fixed Costs	Don't change w/ volume (rent, salaries, software)
Variable Costs	Scale w/ volume (materials, packaging, delivery)

$\text{Total Cost} = \text{Fixed} + (\text{Variable/unit} \times \text{Units})$

Projected Revenue asks: WHO buys? WHAT? At what PRICE? Use **bottom-up**, avoid "1% of market" fallacy.

L9 · Financial Statements & Ratios ★

Income (P&L)	Revenue – Costs = Profit over a period
Balance Sheet	Assets = Liabilities + Equity (snapshot)
Cash Flow	Actual cash in/out (≠ profit!)

$\text{Gross Profit} = \text{Revenue} - \text{COGS}$

$\text{Net Profit} = \text{Gross Profit} - \text{OpEx/tax/interest}$

$\text{Breakeven Units} = \text{Fixed} \div (\text{Price} - \text{Variable/unit})$

$\text{Contribution Margin} = \text{Price} - \text{Variable/unit}$

$\text{Profit Margin} = \text{Net Profit} \div \text{Revenue} \times 100$

$\text{ROI} = (\text{Gain} - \text{Cost}) \div \text{Cost} \times 100$

$\text{Current Ratio} = \text{Current Assets} \div \text{Current Liab.}$

⚡ Profitable ≠ cash-positive — timing of cash flows matters. Burn rate = how fast reserves are spent.

L1 - Why Startups Fail + HCD

Fail: build for selves not users; start with solution not problem; same-background teams → same blind spots.

Bad: Idea→Build→Hope. Good: **People→Problem→Solution.**

Human-Centered Design = start with people's frustrations, needs, behavior. "Users describe solutions; designers discover needs."

⚡ DT = "Observation without Judgment." Your expertise is also a filter that hides reality.

L1 - Cognitive Biases

Halo Effect	One great feature blinds you to flaws
Similarity (Like-Me)	Assume users behave like you
Confirmation	Seek info that supports your idea
Stereotyping	Over-generalize user groups

L1 - Human Hacks & Journey

Human Hack = non-standard workaround because a system failed a need. **Where there's a workaround, there's an unmet need.** (infra/design/trust/process failure)

Bootcamp journey:

Understand→Research→Define→Ideate→Prototype→Test→Evaluate→Pitch

"Business = destination. Design = the maze. Don't jump over it — walk through."

Decode complaints: "slides earlier" → control; "hate group projects" → fair workload; "bigger battery" → anxiety reduction.

Always go one level deeper.

L2 - Observe · Engage · Immerse

Self-reporting fails: people don't do what they say / think / are told.

Observe	Watch in context — habits, workarounds, friction, hesitation. Don't intervene.
Engage	Ask behavioral Qs (not opinions). Decode: checks phone=boredom, avoids speaking=discomfort.
Immerse	Experience it yourself → how it FEELS. <i>Write with non-dominant hand.</i>

⚡ Observed behavior > stated preference. Always.

L2 - Research Types

Primary (direct): observe, behavioral interviews, immersion.

Secondary (indirect): reports, stats, articles, existing data. Replaces assumptions w/ evidence.

Every insight = **User + Need + Context.**

"You think you understand a problem — you saw a symptom."

L3 - PESTEL + System Thinking

Business sees the MAP; Design sees the TRAP (what blocks people).

PEST	(as PEST) policies, economy, social, tech
Environmental	Sustainability, climate, regulations
Legal	Employment, consumer protection, IP laws

System Thinking: user lives in a network. You sell to their father/friends/boss/system too. If user says YES but environment says NO → idea is dead.

L3 - Jobs-to-be-Done + Kano

JTBD — people "hire" products for a job:

- **Functional** — the practical task
- **Social** — how they appear to others
- **Emotional** — how they feel

Not a drill — a hole to hang a photo to feel like a good parent.

Kano (value layers): Required (must-have) · Expected (baseline) · Desired (nice) · Unexpected (delight). Delights drift → become required over time.

L3 - Problem Statement Formula ★

[User] struggles with [situation] because [root cause], leading to [consequence].

Good: "Busy students skip meals at peak hours because food queues are unpredictable, leading to fatigue + lower performance."

Bad: "Students need better campus food" → vague, no root cause.

L4 - Personas & Beachhead

Myth of "everyone": if customer = everyone, you have no customer. "Students/People/Users" = categories, not segments.

Real user = specific situation + specific action + real struggle. "If you can't see the moment, you don't understand the problem."

Persona parts: When (situation) · What they do (behavior) · Avoid (pains) · Feel (emotional job).

Beachhead persona = Segment + real human + real moment + job + intensity.

⚡ Shark bite (urgent, act NOW) vs Mosquito bite (annoying, ignored). Beachhead needs a SHARK-bite problem.

L4 - Root Cause & Jargon

5 Whys → drill to root cause. Move "What is" → "What if." Defined the cage (problem)? Now find the exit (solution).

Jargon Assassin: jargon = symptom of unclear thinking. Explain to a 10-yr-old / grandmother in Amman. **"Speak Human, Not Corporate."**

L5 · How Might We (HMW) — 4 Types ★

HMW = decision about WHICH part of problem space to explore. Template: "HMW help [user] achieve [job] despite [pain]?"

1 Efficiency	"reduce waiting time" → faster/optimize
2 Experience	"reduce stress while waiting" → emotional
3 Behavior	"change WHEN students eat" → new habits
4 System	"redesign food distribution" → structural

⚡ The HMW type you choose determines the kind of solutions you'll get.

L5 · Divergence & Ideation

First idea = obvious. Generate MANY **fundamentally different** ideas. Rules: quantity over quality, no judging, different directions.

Wrong (same thinking): app w/ notifications, app w/ tracking, app w/ AI.

Right (diff thinking): eliminate queues, shift eating times, remove need for cafeteria.

Without HMW: different but irrelevant. With HMW: different AND relevant.

Communication: fillers (umm, like) = panic. The Pause = confidence. Slow down at "the because" (the insight).

L6 · Validation Engine + Double Diamond

Insight→Hypothesis→Prototype→Test→Learn→Decide

Double Diamond: Discover (expand) → Define (focus) → Develop (expand) → Deliver (focus). **Define = hardest step** (constructing the right problem). Develop = HMW zone.

⚡ Fatal mistake: testing HOW (usability) before WHY (value). Confirm they want it first.

L6 · Prototype Types

Storyboard	Tests journey/logic (6–8 steps + emotion). Can't test demand/interaction.
Role-Play	Tests service/conversation. Depends on acting.
Interface (paper/digital)	Tests screens/flow — watch wrong taps, hesitation.
Digital MVP	Real interaction, 1 screen/flow/function. Risk: overbuilding.
Landing/Fake Door	Tests DEMAND — "sign up" → watch clicks. Tests intent, not behavior.

Outdoor signals: "I would use this" → ignore (cheap talk); "Interesting but..." → dig; "When can I get it?" → real demand. Patterns > opinions.

L7 · Feedback Extraction

Feedback = **extraction** of behavior, friction, expectation gaps, commitment. You tested assumptions, not ideas. Testing w/o change = performance.

Evidence Chain: Observation → Insight → Decision → Alignment. Skip one = guessing.

"If you can't say this, you didn't learn": We observed ____ · This means ____ · So we changed ____ · This aligns with the user because ____.

L7 · Feedback Methods + RAT

Silent Observation	No help/guidance — watch hesitation, breakdown
Thinking Aloud	User narrates → where mental model breaks
Wizard of Oz	Simulate value manually w/o building tech
Commitment Test	Push for action/effort. Refusal invalidates prior praise.

RAT = Riskiest Assumption Test: "if THIS is wrong → everything collapses." Define success rate AND failure threshold before testing. Test only with your persona. Never ask "Do you like it?"

L7 · Iterate / Pivot / Kill ★

Usability Friction (HOW)	Wants value but blocked → Iterate (fix flow)
Value Deficit (WHY)	Understands but doesn't care → Pivot (weak problem)
Mental Model Mismatch	Uses it differently → Investigate (new opportunity)

Decision Layer: Iterate = real problem, weak solution · Pivot = wrong problem · Kill = no behavioral intent after multiple tests.

Cost of being wrong: MVP (fast, controlled) → Pilot (slow, limited) → Open Beta (can't hide) → Polished Launch (danger). "Launch when you can't iterate = gambling."

⚡ EXAM QUICK-REFERENCE & COMMON TRAPS

📌 Formula Bank (all math in one place)

Total Cost = Fixed + (Variable/unit × Units)

Contribution Margin (CM) = Price – Variable/unit

Breakeven Units = Fixed Costs ÷ CM

Breakeven Revenue = Breakeven Units × Price

Gross Profit = Revenue – COGS

Net Profit = Gross Profit – OpEx – tax – interest

Gross Profit Margin = Gross Profit ÷ Revenue × 100

Net Profit Margin = Net Profit ÷ Revenue × 100

ROI = (Gain – Cost) ÷ Cost × 100 = Net Profit ÷ Invested × 100

Current Ratio = Current Assets ÷ Current Liabilities

SAM = TAM × target segment %

SOM = SAM × market share (conversion) %

📊 Worked Example — Breakeven

Café: Fixed = 40,000 · Price = 4 · Variable = 1.50 · Sold = 12,000 cups.

- CM = 4 – 1.50 = **2.50/cup**
- Total contribution = 12,000 × 2.50 = **30,000**
- Result = 30,000 – 40,000 = **-10,000 (loss)**
- Breakeven = 40,000 ÷ 2.50 = **16,000 cups**

⚡ Compare total contribution to fixed costs — don't compare revenue to fixed costs.

📊 Worked Example — TAM/SAM/SOM

3M smartphone users · 60% health-conscious · 25% willing to pay · 10% Year-1 conversion.

- TAM = **3,000,000** (all users)
- SAM = 1,800,000 × 25% = **450,000**
- SOM = 450,000 × 10% = **45,000**

⚡ Raise price → fewer "willing to pay" → SAM shrinks → SOM shrinks. Annual revenue = users × monthly price × 12.

⚠️ Most-Confused Distinctions

PEST vs PESTEL	PESTEL = PEST + Environmental + Legal
Fixed vs Variable	Fixed = same regardless of volume; Variable = scales w/ units
CapEx vs OpEx	CapEx = long-term asset (oven); OpEx = day-to-day running cost
Penetration vs Skimming	Low price to grab share vs High price first then lower
Pain vs Gain	Pain = avoid (frustration); Gain = desired positive outcome
Strength vs Opportunity	Strength = internal +; Opportunity = external +
Weakness vs Threat	Weakness = internal –; Threat = external –
Profit vs Cash Flow	Can be profitable but cash-broke (timing of payments)

⚠️ Margin Reclassification Traps

- Move cost **COGS** → **OpEx**: Gross Profit ↑ (GPM ↑), Net Profit unchanged (NPM same).
- Move cost **OpEx** → **COGS**: Gross Profit ↓ (GPM ↓), NPM same.
- Add **depreciation to OpEx**: NPM ↓, GPM unchanged.
- Same Net Profit on **lower revenue** = higher NPM.

⚡ Moving a cost between lines never changes Net Profit — only where the cut appears.

🎯 Decision Frameworks at a Glance

Iterate	Real problem, weak solution (fix HOW)
Pivot	Wrong/weak problem (the WHY)
Kill	No behavioral intent after multiple tests
Desirable	People want it (user)
Feasible	Tech can build it
Viable	Business sustains it (money)

Adoption mapping: SOM = Innovators (2.5%) + Early Adopters (13.5%). SAM extends through Early Majority.

🔑 High-Yield One-Liners

- Innovation must create **value**, not just be invented.
- No defined segment = **no customer**.
- Observed behavior > stated preference.
- Test **WHY (value)** before **HOW (usability)**.
- "I would use this" = cheap talk; "When can I get it?" = real demand.
- Beachhead needs a **shark-bite** (urgent) problem.
- Define = hardest step (construct the right problem).
- If user says YES but their system says NO → idea dies.